

Florence: Infrastructure - Shorebird

1. Overview and Purpose

Shorebird provides Over-The-Air (OTA) update capabilities for the Florence Flutter application. It allows the development team to push critical bug fixes, UI tweaks and performance improvements directly to user devices without requiring a full release.

2. Configuration and Secrets Management

Environment Variable/Secret	Description	Where it is stored	Required By
SHOREBIRD_TOKEN	Authentication token for the Shorebird CLI.	GitHub Secrets	GitHub Actions CI/CD
app_id	Unique identifier for the Florence app (518a6fe4-4e94-4d82-9984-b2f513f9f78a).	shorebird.yaml (Source Code)	Flutter App, Shorebird CLI

3. Technical Implementation Details

- **App Configuration:** The application is registered in Shorebird with the designated App ID. The shorebird.yaml file is committed to version control to ensure the build pipeline can correctly identify the target application during compilation.
- **Automatic Updates:** The auto_update feature is currently enabled by default. When a user launches the Florence app it will silently check for and download available patches in the background ensuring users always have the latest stable code.
- **Release vs Patch Strategy:**
 - **Full Releases:** When the version in pubspec.yaml is incremented, the CI/CD pipeline executes shorebird release android to compile a new binary and upload it to Firebase App Distribution.

- **OTA Patches:** When code is merged to the production branch without a version bump the pipeline executes shorebird patch android . This pushes the Dart and Flutter changes directly to the Shorebird servers.
- **Tracks:** Patches are published to the default stable track. Shorebird also supports beta or custom tracks which can be utilised in the future for percentage-based rollouts or internal testing groups (pro plan might be required for this).
- **Flutter Engine:** The application is currently compiled against Flutter version 3.38.4.

4. Billing Limits and Day 2 Operations

- **Current Tier:** Free
- **Patching Constraints:** OTA patches can only modify Dart and Flutter code. They cannot modify native Android or iOS code, add new native dependencies to pubspec.yaml or change the underlying Flutter engine version. If native changes or new third-party packages are required, a full release must be triggered by bumping the version number.
- **Day 2 Operations:** To push a hotfix the development team simply merges the corrected Dart code into the production branch. The GitHub Actions workflow will automatically detect the lack of a version change and deploy the patch via Shorebird.